

A Thermodynamically-Consistent MeshGraphNet Surrogate for Transient Welding Heat Transfer: Enthalpy-State GENERIC Structure and Push-Forward Training

Abstract

We develop a graph-neural-network surrogate for transient thermal welding simulation that replaces an expensive nonlinear finite-element (FEM) solver with a fast, autoregressive learned model. The surrogate follows the Encoder–Processor–Decoder *MeshGraphNet* (MPNN) architecture and is trained on snapshot pairs produced by a 2D transient heat-conduction solver with a moving Goldak double-ellipsoid arc source, temperature-dependent apparent heat capacity (latent heats of fusion and vaporization), and mixed convective/radiative/Dirichlet boundaries. We identify and correct a subtle but consequential thermodynamic inconsistency that arises when one imposes GENERIC (*General Equation for the Non-Equilibrium Reversible–Irreversible Coupling*) structure directly on temperature: under strong latent heat, conserving the sum of nodal temperatures is *not* conserving energy. The fix is the classical *enthalpy method*: carry the volumetric enthalpy as the conserved state, evolve it with a learned symmetric positive semi-definite (SPSD) graph-Laplacian dissipation operator, and recover temperature by inverting the latent-heat-aware enthalpy curve. Taking the *extensive* nodal energy as the conserved state makes this guarantee exact on arbitrary (including non-uniform) meshes, and we treat the arc source and boundary losses through the open-system extension of GENERIC. This guarantees the first and second laws *by construction*, even through phase change. A learned, per-node *enriched source* restores the melt-pool peak heating that the structure-preserving dissipation alone cannot supply, without breaking the guarantees. Finally, we align the training objective with long autoregressive rollouts using *push-forward* ($K=2$) multi-step backpropagation. On a held-out, out-of-distribution Archimedean-spiral weld (1526 autoregressive steps), the model attains a global RMSE of 33.2 K while respecting the maximum principle (no sub-ambient cooling) and remaining stable. A matched ablation (GENERIC enthalpy structure ON vs. OFF, both at $K=2$) shows that the thermodynamic structure provides $\sim 3\times$ lower rollout-RMSE variance, maximum-principle respect, and a validation metric that tracks the long rollout, whereas the unconstrained network is accurate-but-fragile.

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1 Introduction

Arc welding involves a moving, highly localized heat source that drives a metal plate through melting and partial vaporization, leaving steep transient thermal gradients in its wake. Accurate prediction of the temperature history is essential for estimating residual stresses, distortion, and microstructure. The standard tool is the transient nonlinear finite-element (FEM) method, but a single high-fidelity simulation can take minutes to hours, making design-space exploration, uncertainty quantification, and online control prohibitively expensive.

Scientific machine learning (SciML) offers a route to fast surrogates that learn the simulator’s input–output map. Among mesh-based learned simulators, *MeshGraphNets* [1] are particularly

attractive: they operate directly on the simulation mesh as a graph, generalize across mesh geometries, and roll forward autoregressively. However, two challenges are acute for welding:

1. **Long-horizon stability.** Autoregressive surrogates accumulate error step over step. A model that is accurate on short validation rollouts can diverge catastrophically (spurious far-field heating, sub-ambient cooling, or global runaway) over the much longer inference horizons of interest.
2. **Phase-change physics.** Welding energy storage is dominated by *latent heat*: melting and boiling absorb large amounts of energy at nearly constant temperature, so the energy–temperature relationship is strongly nonlinear. A surrogate that implicitly assumes $\text{energy} \propto \text{temperature}$ mis-handles exactly the melt pool, which is the region of greatest interest.

This paper addresses both. We build a configurable MeshGraphNet surrogate and equip it with an optional *structure-preserving thermodynamic head* derived from the GENERIC framework [3, 4]. Our contributions are:

1. A complete, reproducible pipeline: a nonlinear 2D transient welding FEM solver (Goldak source, apparent heat capacity, Robin/Dirichlet boundaries, θ -method with Picard iteration), a physics-aware graph representation with purely *relative* features, and an Encoder–Processor–Decoder MPNN (Sections 2, 4 and 5).
2. A diagnosis of the thermodynamic inconsistency of imposing GENERIC on temperature under latent heat, and its correction via an *enthalpy-state* GENERIC head that conserves energy and produces entropy by construction, with a numerically well-conditioned temperature-equivalent rescaling (Sections 6 and 7). Choosing the *extensive* nodal energy $\mathcal{E}_i = h_i V_i$ as the state makes energy conservation exact on arbitrary (non-uniform) meshes, and the arc/boundary terms are handled by the open-system extension of GENERIC.
3. An *enriched source* that restores melt-pool peak heating while leaving the dissipative operator structure-preserving (Section 8).
4. A *push-forward* multi-step training scheme that aligns the objective with long rollouts (Section 9), and a matched ablation isolating the value of the thermodynamic structure (Section 10).

2 Governing physics and the finite-element solver

2.1 Continuum model

We model the top-down (in-plane) surface of a thin plate of thickness H_z , and solve the through-thickness-integrated transient heat-conduction equation for the temperature field $T(\mathbf{x}, t)$, $\mathbf{x} \in \Omega \subset \mathbb{R}^2$:

$$\rho c_p^{\text{app}}(T) \frac{\partial T}{\partial t} = \nabla \cdot (k(T) \nabla T) + Q(\mathbf{x}, t), \quad (1)$$

where ρ is density, k thermal conductivity, Q the volumetric heat source, and $c_p^{\text{app}}(T)$ the *apparent* heat capacity, which folds latent heat into an effective specific heat (Section 2.2). Boundaries $\partial\Omega$ carry either a Dirichlet condition $T = T_D$ or a Robin condition combining convection and (optionally linearized) radiation,

$$-k \nabla T \cdot \mathbf{n} = h_{\text{conv}} (T - T_\infty) + \varepsilon \sigma_{\text{SB}} (T^4 - T_\infty^4), \quad (2)$$

with outward normal $\mathbf{n}$, convection coefficient h_{conv} , ambient T_∞ , emissivity ε , and Stefan–Boltzmann constant $\sigma_{\text{SB}} = 5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$.

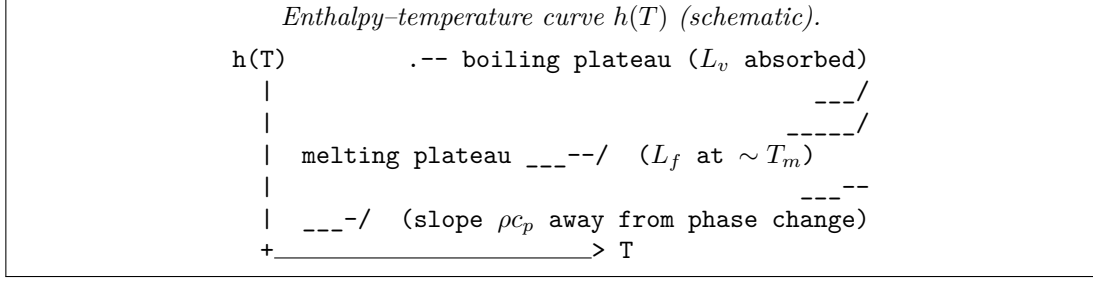


Figure 1: The enthalpy–temperature relation $h(T)$. Near melting and boiling a large change in enthalpy (energy) yields almost no change in temperature: the phase change absorbs the energy. This nonlinearity is what a temperature-state GENERIC head mis-models and what the enthalpy-state head captures exactly.

2.2 Latent heat and the enthalpy curve

The defining physical feature of welding is that the metal melts (and locally boils): a phase change absorbs a large *latent heat* at nearly constant temperature, so c_p^{app} is strongly temperature-dependent. We model it as a sensible heat capacity plus two normalized Gaussian bumps centered on melting and boiling:

$$c_p^{\text{app}}(T) = c_p^{\text{sens}} + L_f \frac{df_\ell}{dT} + L_v \frac{df_v}{dT}, \quad (3)$$

$$\frac{df_\ell}{dT} = \frac{1}{s_\ell \sqrt{\pi}} e^{-((T-T_m)/s_\ell)^2}, \quad \frac{df_v}{dT} = \frac{1}{s_v \sqrt{\pi}} e^{-((T-T_b)/s_v)^2}, \quad (4)$$

where L_f, L_v are the latent heats of fusion and vaporization, $T_m = \frac{1}{2}(T_{\text{sol}} + T_{\text{liq}})$ is the mean melting temperature, T_b the boiling point, and s_ℓ, s_v smoothing widths. Because each bump integrates to one, $\int c_p^{\text{app}} dT$ recovers exactly L_f and L_v . The vaporization bump acts as a physical *thermostat*: energy that would otherwise superheat near-pool nodes to unphysical values ($\sim 10^4$ K) is instead absorbed by the (smeared) phase change, capping the dynamic range near boiling.

Integrating Eq. (3) gives the *volumetric enthalpy* (energy stored per unit volume relative to a reference T_0):

$$h(T) = \rho \int_{T_0}^T c_p^{\text{app}}(\tau) d\tau. \quad (5)$$

Away from phase change $h(T) \approx \rho c_p^{\text{sens}} (T - T_0)$ is nearly linear; across melting/boiling the curve develops plateaus where a large change in h produces an almost-zero change in T (equivalently the slope $dh/dT = \rho c_p^{\text{app}}$ spikes). This curve, illustrated schematically in Fig. 1, is the crux of the thermodynamic argument in Section 6. Numerical parameters (mild steel) are listed in Table 1.

2.3 Goldak double-ellipsoid heat source

The arc is modeled by the Goldak double-ellipsoid distribution [6]. For the thickness-integrated 2D model we analytically integrate the 3D ellipsoid through the plate thickness to obtain a surface flux, then divide by H_z to get a volumetric source. With the torch at position $\mathbf{p}(t)$ moving along trajectory $\boldsymbol{\gamma}(s)$ with unit tangent $\hat{\mathbf{t}}$ and normal $\hat{\mathbf{n}}$, define the co-moving coordinates $\xi = (\mathbf{x} - \mathbf{p}) \cdot \hat{\mathbf{t}}$ (along travel) and $\eta = (\mathbf{x} - \mathbf{p}) \cdot \hat{\mathbf{n}}$ (transverse). Then

$$Q(\mathbf{x}, t) = \frac{1}{H_z} \frac{6 f \eta_{\text{eff}} P}{\pi b c} \exp\left(-\frac{3\xi^2}{c^2} - \frac{3\eta^2}{b^2}\right), \quad (6)$$

Table 1: Material and source parameters (mild-steel defaults, SI/kelvin).

Symbol	Quantity	Value
ρ	density	7850 kg m^{-3}
c_p^{sens}	sensible specific heat	$600 \text{ J kg}^{-1} \text{ K}^{-1}$
k	thermal conductivity	$30 \text{ W m}^{-1} \text{ K}^{-1}$
L_f	latent heat of fusion	$2.7 \times 10^5 \text{ J kg}^{-1}$
$T_{\text{sol}}, T_{\text{liq}}$	solidus, liquidus	1723 K, 1773 K
L_v	latent heat of vaporization	$6.3 \times 10^6 \text{ J kg}^{-1}$
T_b	boiling point	3134 K
s_v (vap_width)	vaporization smoothing	120 K
H_z	plate thickness	4 mm to 8 mm
P	gross arc power	1500 W to 3500 W
η	arc efficiency	0.70 to 0.90

where $\eta_{\text{eff}}P$ is the net deposited power (η_{eff} the arc efficiency), b is the transverse half-width, and the along-travel semi-axis and fraction switch between the front lobe ($c = c_f$, $f = f_f$ for $\xi \geq 0$) and the rear lobe ($c = c_r$, $f = f_r$ for $\xi < 0$), with the Goldak constraint $f_f + f_r = 2$. During a post-weld cooling phase the source is switched off ($Q \equiv 0$) and the torch is parked at its final weld position.

2.4 Discretization and time integration

We discretize Eq. (1) with continuous piecewise-linear (P_1) triangular finite elements (`scikit-fem` [9]). The weak form yields the semidiscrete system

$$\mathbf{M}(T) \dot{\mathbf{T}} + \mathbf{A}(T) \mathbf{T} = \mathbf{b}(t), \quad (7)$$

where $\mathbf{M}(T)$ is the (temperature-dependent) heat-capacity mass matrix with entries $\int_{\Omega} \rho c_p^{\text{app}}(T) \phi_i \phi_j$, $\mathbf{A} = \mathbf{K} + \mathbf{R}$ collects the conduction stiffness $\mathbf{K} = \int_{\Omega} k \nabla \phi_i \cdot \nabla \phi_j$ and the Robin convection/radiation operator $\mathbf{R}$, and $\mathbf{b}$ is the Goldak load plus boundary loads. Time integration uses the θ -method (default backward Euler, $\theta = 1$),

$$\left(\frac{1}{\Delta t} \mathbf{M} + \theta \mathbf{A} \right) \mathbf{T}^{n+1} = \frac{1}{\Delta t} \mathbf{M} \mathbf{T}^n - (1 - \theta) \mathbf{A} \mathbf{T}^n + \theta \mathbf{b}^{n+1} + (1 - \theta) \mathbf{b}^n. \quad (8)$$

Because $c_p^{\text{app}}(T)$ (latent heat) and radiation make the problem nonlinear, each step is solved by a Picard fixed-point iteration (re-assembling $\mathbf{M}$ and the linearized radiation operator $\mathbf{R}_{\text{rad}} = \varepsilon \sigma_{\text{SB}}(T^2 + T_{\infty}^2)(T + T_{\infty})$ at the current iterate) until the relative update falls below a tolerance. Dirichlet constraints are imposed by condensation. The solver records a time series of snapshots $\{\mathbf{T}^n\}$ together with the torch frame and source state, and serializes everything (including all process/boundary metadata) so the graph pipeline can reconstruct every feature from a single self-contained file.

3 Dataset generation

We generate a corpus designed so that a single generalization axis—the weld *trajectory topology*—is held out, while every other axis (geometry, plate size, process parameters, boundary conditions) is covered by the training distribution. This makes any generalization gap cleanly attributable to the trajectory rather than to extrapolation in plate size or speed.

Sampling ranges. Each simulation draws i.i.d. from: plate bounding box 0.08 m to 0.25 m per side; element edge ≈ 3 mm (yielding ~ 800 to 6600-node meshes); gross power 1500 W to 3500 W; arc efficiency 0.70 to 0.90; travel speed 2 mm/s to 12 mm/s; Goldak $b, c_f \in 2.0$ mm to 3.5 mm, $c_r/c_f \in [1.6, 2.4]$, $f_f \in [0.5, 0.7]$; thickness 4 mm to 8 mm; ambient 290 K to 320 K; convection $h_{\text{conv}} \in 5 \text{ W m}^{-2} \text{ K}^{-1}$ to $40 \text{ W m}^{-2} \text{ K}^{-1}$; emissivity in $\{0, 0.2, 0.35\}$ (weighted toward zero), with a 30% chance that one outer edge is held Dirichlet. Boundary-condition strata (radiation $\times$ Dirichlet) are cycled to guarantee balanced coverage. Geometries are rectangular (50%), L-shaped (25%), or holed (25%); trajectories are straight, diagonal, sinusoidal, or circular-arc paths.

Adaptive cooling tail. After the weld phase, the source is switched off and the simulation continues until the peak excess over ambient falls below 7% of the welding peak (near-field relaxed) or a cap of 30 s is reached. This guarantees every trajectory includes a complete local cool-down, which proved essential for suppressing spurious far-field heating in long rollouts. The time step is pinned near $\Delta t \approx 0.05$ s (matching the inference regime).

Corpus. The final dataset is 360 simulations (300 train / 60 validation), 398 597 snapshot-pair graphs in total. Splitting is done at the *simulation* level (every snapshot of a given weld lands in one fold) to prevent temporal leakage of adjacent states across folds.

4 Graph representation

Each consecutive snapshot pair (T^t, T^{t+1}) of a simulation becomes one graph. The mesh nodes are graph nodes; mesh edges (both orientations, no self-loops) are graph edges. The learning target is the per-node temperature increment $\mathbf{y} = T^{t+1} - T^t \in \mathbb{R}^N$.

A central design choice is to use *only relative* features—no absolute coordinates—so the model cannot memorize positions and must learn local physics:

Node features (12-d).

$[T, q_{\text{Goldak}}, \xi', \eta', \eta_{\text{eff}}P, v, \mathbb{K}_{\text{int}}, \mathbb{K}_{\text{Dir}}, \mathbb{K}_{\text{Rob}}, h_{\text{conv}}, \varepsilon, T_{\infty}]$. Here q_{Goldak} is the analytical source Eq. (6) evaluated at the node (gated to zero during the cooling tail); (ξ', η') are the co-moving torch coordinates in the $(\hat{\mathbf{t}}, \hat{\mathbf{n}})$ frame; $\eta_{\text{eff}}P$ and v are process scalars broadcast to all nodes; the three one-hots encode the node’s boundary type; and $h_{\text{conv}}, \varepsilon, T_{\infty}$ are the local boundary values. The Goldak semi-axes are *not* exposed as features: they are global per-simulation constants already folded into q_{Goldak} and (ξ', η') .

Edge features (3-d).

$[\Delta x, \Delta y, \|\mathbf{u}_{ij}\|]$ with $\mathbf{u}_{ij} = \mathbf{x}_i - \mathbf{x}_j$. Geometry enters only through these displacement vectors.

Continuous features and the target are standardized by z-score using statistics computed over the training corpus; the three one-hot columns are left untouched. Crucially, normalization constants are stored with the model so inference reproduces the exact transform.

5 MeshGraphNet architecture (the MPNN)

The backbone is the Encoder–Processor–Decoder MeshGraphNet of Pfaff et al. [1], a message-passing neural network (MPNN). Let $\mathbf{x}_i \in \mathbb{R}^{12}$ be node features and $\mathbf{e}_{ij} \in \mathbb{R}^3$ edge features. All sub-networks are multilayer perceptrons (MLPs) of width $d = 128$ with two hidden layers, ReLU activations, and a trailing LayerNorm (except the decoder).

Encoder. Two independent MLPs lift raw features into a shared d -dimensional latent space:

$$\mathbf{v}_i^{(0)} = \text{MLP}_{\text{enc}}^v(\mathbf{x}_i), \quad \boldsymbol{\epsilon}_{ij}^{(0)} = \text{MLP}_{\text{enc}}^e(\mathbf{e}_{ij}). \quad (9)$$

Processor. A stack of L GraphNet blocks (here $L = 8$). Each block updates edges from their endpoints, aggregates messages at receiver nodes, and updates nodes, all with *residual* connections that let gradients survive the deep stack (important for backpropagating through multi-step rollouts):

$$\boldsymbol{\epsilon}_{ij}^{(\ell+1)} = \boldsymbol{\epsilon}_{ij}^{(\ell)} + \text{MLP}_e^{(\ell)}([\boldsymbol{\epsilon}_{ij}^{(\ell)}, \mathbf{v}_i^{(\ell)}, \mathbf{v}_j^{(\ell)}]), \quad (10)$$

$$\mathbf{m}_j^{(\ell+1)} = \sum_{i \in \mathcal{N}(j)} \boldsymbol{\epsilon}_{ij}^{(\ell+1)}, \quad (\text{sum aggregation}) \quad (11)$$

$$\mathbf{v}_j^{(\ell+1)} = \mathbf{v}_j^{(\ell)} + \text{MLP}_v^{(\ell)}([\mathbf{v}_j^{(\ell)}, \mathbf{m}_j^{(\ell+1)}]). \quad (12)$$

With $L = 8$ message-passing hops, information propagates up to 8 mesh edges per step, giving the receptive field needed to represent the heat-affected zone around the torch.

Decoder. A final MLP (no LayerNorm, so it can emit unbounded physical values) projects node latents to the scalar output:

$$\widetilde{\Delta T}_i = \text{MLP}_{\text{dec}}(\mathbf{v}_i^{(L)}) \in \mathbb{R}. \quad (13)$$

In the *baseline* (unconstrained) model this normalized increment is the prediction. In the structure-preserving variants the decoder is bypassed and the increment is built instead from the processor latents through the thermodynamic head (Sections 6 and 7). The configuration used throughout has $d = 128$, $L = 8$, $\approx 1.29\text{M}$ parameters for the unconstrained model.

Inference (autoregressive rollout). At test time the model receives only the initial field T^0 and the (known) Goldak schedule. Because the trajectory is deterministic, all time-dependent features (q_{Goldak} , (ξ', η') , v) are precomputed for every step; during the loop only the temperature column is overwritten with the running prediction:

$$T^{t+1} = T^t + \mathcal{N}_y^{-1}(f_\theta(\mathbf{x}^t; \text{edges})), \quad (14)$$

where $\mathcal{N}_y^{-1}$ de-normalizes the predicted ΔT . The same feature construction code is shared between training and rollout to guarantee parity.

6 Structure-preserving thermodynamics: the GENERIC head

The unconstrained MPNN is accurate but can violate physics over long rollouts (spurious global heating, sub-ambient cooling, runaway). We therefore optionally replace the free decoder increment by an increment that satisfies the laws of thermodynamics *by construction*, using the GENERIC framework [3, 4].

6.1 GENERIC in a nutshell

GENERIC writes the evolution of a state $\mathbf{z}$ as a reversible plus a dissipative part, driven by two potentials—total energy $E(\mathbf{z})$ and entropy $S(\mathbf{z})$:

$$\frac{d\mathbf{z}}{dt} = \underbrace{\mathbf{L}(\mathbf{z}) \frac{\partial E}{\partial \mathbf{z}}}_{\text{reversible}} + \underbrace{\mathbf{M}(\mathbf{z}) \frac{\partial S}{\partial \mathbf{z}}}_{\text{dissipative}}, \quad (15)$$

with $\mathbf{L}$ antisymmetric, $\mathbf{M}$ symmetric positive semi-definite (SPSD), and the *degeneracy conditions*

$$\mathbf{L} \frac{\partial S}{\partial \mathbf{z}} = \mathbf{0}, \quad \mathbf{M} \frac{\partial E}{\partial \mathbf{z}} = \mathbf{0}. \quad (16)$$

These immediately yield the two laws:

$$\frac{dE}{dt} = 0 \quad (\text{1st law}), \quad \frac{dS}{dt} = \left(\frac{\partial S}{\partial \mathbf{z}} \right)^\top \mathbf{M} \left(\frac{\partial S}{\partial \mathbf{z}} \right) \geq 0 \quad (\text{2nd law}). \quad (17)$$

Pure heat conduction is *purely dissipative*: there is no reversible heat flow, so $\mathbf{L} = \mathbf{0}$ and $\dot{\mathbf{z}} = \mathbf{M} \partial S / \partial \mathbf{z}$ (plus a non-conservative exchange with the external source/environment). Two choices then decide physical consistency: *what is the state $\mathbf{z}$, and is $\mathbf{M} \partial E / \partial \mathbf{z} = \mathbf{0}$ the correct energy conservation.*

Open, driven systems. The closed-form identities Eq. (17) describe an *isolated* system; welding is *open and driven*—the arc injects energy through the Goldak source and the boundaries exchange heat by convection and radiation. We therefore use the standard boundary/source extension of GENERIC for open systems [4]: the structure-preserving operator acts only on the *internal* bulk evolution, and the exchange with the environment is added as a separate, non-conservative term Δ^{ext} evaluated after the internal dissipative step [5, 4], in which boundary and bulk contributions to the brackets are separated so that the isolated-system structure of Eq. (17) is recovered for the bulk while the boundary/source terms account for the exchange with the environment,

$$\dot{\mathbf{z}} = \underbrace{\mathbf{M} \partial S / \partial \mathbf{z}}_{\text{internal (degeneracy + 2nd law)}} + \underbrace{\mathbf{f}^{\text{ext}}}_{\text{source + boundary fluxes}}. \quad (18)$$

Consequently the degeneracy and second-law guarantees attach to the *internal* operator: $\mathbf{1}^\top \mathbf{M} \partial S / \partial \mathbf{z} = 0$ states that the irreversible bulk dynamics conserves energy, while the total energy changes by *exactly* the energy crossing the boundary or deposited by the source (as physics demands, not a defect); and $(\partial S / \partial \mathbf{z})^\top \mathbf{M} (\partial S / \partial \mathbf{z}) \geq 0$ is the non-negative *internal entropy production*, the correct second-law statement for an open system (the system’s total entropy may still fall as heat leaves through the boundary). The external term $\mathbf{f}^{\text{ext}}$ is realized by the (enriched) Goldak source and Newton/Stefan cooling of Section 8; it is the only part not constrained by the GENERIC structure, by design.

6.2 A learned SPSP operator on the graph

We realize $\mathbf{M}$ as a *graph Laplacian of learned conductances*. The processor produces, for each undirected edge, a non-negative symmetric weight

$$w_{ij} = e^\alpha \text{softplus} \left(\text{MLP}(\mathbf{v}_i + \mathbf{v}_j, (\mathbf{v}_i - \mathbf{v}_j)^2, T_i + T_j, |T_i - T_j|) \right) = w_{ji} \geq 0, \quad (19)$$

where $\mathbf{v}$ are processor node latents and α a learnable global log-scale (initialized so the dissipation starts at $O(1\text{K})$ per step). Symmetry is obtained *for free* by feeding the MLP only features symmetric under $i \leftrightarrow j$. Defining $\mathbf{W} = [w_{ij}]$, $\mathbf{D} = \text{diag}(\mathbf{W}\mathbf{1})$, the Laplacian is $\mathbf{L}_w = \mathbf{D} - \mathbf{W}$, which is SPSP and satisfies $\mathbf{L}_w \mathbf{1} = \mathbf{0}$ (zero row/column sums).

The dissipative increment acts on the entropy gradient $\boldsymbol{\mu}$, the *thermodynamic force*:

$$\mu_i \equiv \frac{\partial S}{\partial e_i} = \frac{1}{T_i}, \quad (20)$$

which is the definition of (inverse) temperature, $dS = dE/T$, and holds for *any* $c_p(T)$, including latent heat. With $\mathbf{M} = \mathbf{L}_w$,

$$(\mathbf{L}_w \boldsymbol{\mu})_i = \sum_{j \sim i} w_{ij} (\mu_i - \mu_j). \quad (21)$$

A hot node (μ_i small) beside cold neighbors (μ_j large) receives a negative increment—it *loses* energy. Heat flows hot→cold by construction.

6.3 The inconsistency of a temperature state under latent heat

It is tempting to take the state as temperature, $\mathbf{z} = \mathbf{T}$, with constant- c_p potentials $E = \sum_i \rho c_p T_i V_i$ and $S = \sum_i \rho c_p \ln(T_i) V_i$, and to write the temperature increment directly as $\Delta T_i = (\mathbf{L}_w \boldsymbol{\mu})_i$. Then $\sum_i \Delta T_i = \mathbf{1}^\top \mathbf{L}_w \boldsymbol{\mu} = 0$ and one is tempted to call this “energy conservation.” It is not. The thermodynamically correct dissipative law evolves *energy*:

$$\frac{de_i}{dt} = (\mathbf{L}_w \boldsymbol{\mu})_i, \quad \frac{dT_i}{dt} = \frac{1}{\rho c_p^{\text{app}}(T_i)} \frac{de_i}{dt}. \quad (22)$$

Writing $\Delta T_i = (\mathbf{L}_w \boldsymbol{\mu})_i$ directly *drops* the factor $1/(\rho c_p^{\text{app}}(T_i))$. When c_p is constant this is a global rescale that the learned w_{ij} can absorb; but with latent heat c_p^{app} varies by orders of magnitude between cold metal and the melt pool, so

$$\sum_i \Delta T_i = 0 \text{ (temperature-sum)} \neq \sum_i \rho c_p^{\text{app}}(T_i) \Delta T_i = \sum_i \Delta e_i = 0 \text{ (energy)}. \quad (23)$$

The two differ precisely by the $c_p^{\text{app}}(T)$ factor that blows up at the melt pool. Physically, Eq. (22) says dT/dt should be *small* there (incoming energy is swallowed by the phase change); the temperature-state head has no such damping and over-moves the hottest nodes. This explains the empirical observation that the temperature-state structure-preserving head *caps/distorts the peak* while the unconstrained network fits it better.

7 The enthalpy-state GENERIC head

The fix is the classical *enthalpy method* of phase-change heat transfer [7]: take energy (enthalpy) as the conserved state and recover temperature by inverting the latent-heat-aware curve.

7.1 State, potentials, and the two laws

Let the proper conserved state be the *extensive nodal energy* $\mathbf{z} = \boldsymbol{\mathcal{E}}$, $\mathcal{E}_i = h_i V_i$ [J], where $h_i = h(T_i)$ is the volumetric enthalpy (Eq. (5), monotone hence invertible) and V_i is the node’s tributary volume (the diagonal of the lumped FEM mass matrix). The total energy is then $E = \sum_i \mathcal{E}_i$, so $\partial E / \partial \mathcal{E}_i = 1$ and the GENERIC degeneracy

$$\mathbf{M} \frac{\partial E}{\partial \mathbf{z}} = \mathbf{L}_w \mathbf{1} = \mathbf{0} \quad (24)$$

holds *exactly on any mesh* (the graph Laplacian has zero column sums). The entropy gradient is $\mu_i = \partial S / \partial \mathcal{E}_i = 1/T_i$ as before (the definition of temperature, valid for any c_p). The dissipative evolution and temperature recovery are

$$\boxed{\frac{d\mathcal{E}_i}{dt} = (\mathbf{L}_w \boldsymbol{\mu})_i = \sum_{j \sim i} w_{ij} \left(\frac{1}{T_i} - \frac{1}{T_j} \right), \quad T_i = h^{-1}(\mathcal{E}_i / V_i).} \quad (25)$$

Remark (implemented uniform-volume reduction). Choosing the *volumetric* enthalpy h as the state (rather than the extensive $\mathcal{E}$) gives $\partial E/\partial h_i = V_i$, so the degeneracy would require $\mathbf{L}_w \mathbf{V} = \mathbf{0}$ rather than $\mathbf{L}_w \mathbf{1} = \mathbf{0}$; with a standard graph Laplacian these coincide *only* for uniform V_i . The meshes used here are uniform tensor-product triangulations (Section 3), so we take equal interior tributary volumes $V_i \equiv V$: the constant V is absorbed into the learned conductance scale e^α , the head evolves the intensive temperature-equivalent enthalpy H (Section 7.3) directly, and conservation reduces to the unweighted $\sum_i \Delta H_i^{\text{diss}} = 0$. This is identical to the extensive form Eq. (25) in the mesh interior and differs only at boundary/corner nodes (tributary volume $V/2, V/4$), a small residual confined to $\partial\Omega$. The general V_i -weighted form Eq. (25) restores *exact* conservation on arbitrary (e.g. arc-refined) meshes at the cost of a single learning-free weighting; we adopt it in the theory and flag the weighting as an implementation refinement (Section 11).

Proposition 1 (Discrete first and second laws, any mesh). For any non-negative symmetric conductances $w_{ij} = w_{ji} \geq 0$ and any nodal volumes $V_i > 0$, the dissipative update Eq. (25) satisfies, per graph,

$$\frac{dE}{dt} = \sum_i \frac{d\mathcal{E}_i}{dt} = \mathbf{1}^\top \mathbf{L}_w \boldsymbol{\mu} = 0 \quad (\text{energy conserved}), \quad \frac{dS}{dt} = \boldsymbol{\mu}^\top \mathbf{L}_w \boldsymbol{\mu} = \frac{1}{2} \sum_{i,j} w_{ij} (\mu_i - \mu_j)^2 \geq 0. \quad (26)$$

Proof. $\mathbf{1}^\top \mathbf{L}_w = \mathbf{0}$ because $\mathbf{L}_w$ has zero column sums, giving the first law; and $\mathbf{L}_w$ is SPSD because $w_{ij} \geq 0$ and symmetric, so the entropy-production quadratic form is non-negative. Crucially, the conserved quantity is now the *extensive energy* $\mathcal{E} = \sum_i \mathcal{E}_i$ (not the temperature sum, nor the unweighted volumetric enthalpy sum), so this is the genuine first law on *any* mesh, even with latent heat. The uniform-volume implementation (above) conserves $\sum_i \Delta H_i^{\text{diss}} = 0$, which equals $dE/dt = 0$ exactly when V_i is constant (mesh interior) and up to the $O(V/2, V/4)$ boundary-node correction otherwise. $\square$

Latent heat is now handled exactly: at the melt pool, $T = h^{-1}(h)$ is nearly flat (the plateau in Fig. 1), so a large Δh produces almost no ΔT , just as in the FEM.

7.2 Connection to Fourier conduction

To check that Eq. (25) reduces to Fourier conduction, note $\nabla T = -T^2 \nabla(1/T) = -T^2 \nabla \mu$, so the conductive flux is $\mathbf{q} = -k \nabla T = k T^2 \nabla \mu$. Discretizing $\nabla \cdot \mathbf{q}$ on the graph yields exactly the Laplacian form $\mathbf{L}_w \boldsymbol{\mu}$ with edge conductances scaling as $w_{ij} \sim k T_i T_j / \ell_{ij}^2$. The network’s task is to learn these (temperature- and geometry-dependent) conductances.

7.3 Numerical conditioning: temperature-equivalent enthalpy

Raw volumetric enthalpy is $h \sim 10^9 \text{ J m}^{-3}$, so the inverse map has slope $\partial T/\partial h = 1/(\rho c_p^{\text{app}}) \sim 2 \times 10^{-7}$. This factor multiplies every gradient flowing back to the conductances and gains, pushing them below Adam’s $\varepsilon = 10^{-8}$ floor and stalling training (empirically: the loss freezes and diffusion never develops). We therefore work in *temperature-equivalent enthalpy* (units of kelvin):

$$H(T) = \frac{h(T)}{\rho c_p^{\text{sens}}} = \int_{T_0}^T \frac{c_p^{\text{app}}(\tau)}{c_p^{\text{sens}}} d\tau. \quad (27)$$

Away from phase change $H \approx T - T_0$ (unit slope); across melting/boiling the slope spikes (the plateau). Since this is a *constant* rescale of h , energy conservation is unchanged ($\sum_i \Delta H_i^{\text{diss}} = 0 \Leftrightarrow \sum_i \Delta h_i^{\text{diss}} = 0$), but now increments are $O(K)$ and gradients $O(1)$. This single rescaling moved the conductance gradients from $\sim 10^{-13}$ to $\sim 10^{-4}$ and unstuck training from the first epoch.

7.4 Discrete scheme

The curve $H(T)$ is tabulated once from the *same* material model used by the FEM, on a uniform grid $T^{(0)} < \dots < T^{(G-1)}$ ($G = 4096$ from 200 K to 8000 K):

$$H^{(g)} = \sum_{m \leq g} \frac{c_p^{\text{app}}(T^{(m)})}{c_p^{\text{sens}}} \Delta T^{(m)}, \quad (28)$$

which is strictly increasing (hence invertible) since $c_p^{\text{app}} > 0$. The arrays $(T^{(g)}, H^{(g)})$ ride in the model’s `state_dict` so checkpoints reproduce the exact physics. Both $H = \mathcal{H}(T)$ and $T = \mathcal{H}^{-1}(H)$ are evaluated by a differentiable piecewise-linear interpolation (binary search for the bracket, linear blend), whose derivative is the local slope—finite and well defined, and inside the autograd graph. One forward step per node:

$$\Delta H_i^{\text{diss}} = \sum_{j \sim i} w_{ij} (\mu_i - \mu_j), \quad \mu_i = 1/T_i, \quad (29)$$

$$H_i^{\text{new}} = \mathcal{H}(T_i) + \Delta H_i^{\text{diss}} + \Delta H_i^{\text{ext}}, \quad (30)$$

$$\Delta T_i = \mathcal{H}^{-1}(H_i^{\text{new}}) - T_i, \quad (31)$$

and the head returns the normalized ΔT_i , so the rollout loop and data pipeline are unchanged (the network still rolls temperature).

8 The enriched source

Both structure-preserving heads (temperature and enthalpy) were observed to run *cold at the active peak*. The reason is structural: the dissipative operator can *only cool* the hottest node (hot→cold by construction); the *only* term that can heat the peak is the external source. With a single global scalar gain it was too rigid to inject enough energy. The external exchange is, however, exactly the *non-conservative* term that the GENERIC degeneracy and second law do *not* constrain—so we may give it expressivity without breaking any guarantee.

We replace the scalar source/cooling gains by per-node learned modulations:

$$\Delta H_i^{\text{ext}} = \underbrace{\text{softplus}(g_q + \text{MLP}_q(\mathbf{v}_i, q_i, T_i))}_{\text{Goldak heating } \geq 0} q_i + \underbrace{\text{softplus}(g_c + \text{MLP}_c(\mathbf{v}_i, T_i, T_\infty))}_{\text{Newton cooling on Robin nodes } r_i} r_i (T_\infty - T_i), \quad (32)$$

where q_i is the node’s Goldak source feature and r_i the Robin one-hot. The MLP output layers are zero-initialized, so the head *starts identical* to the scalar version and then learns where to inject more energy. The heating gain is $\text{softplus}(\cdot) \geq 0$ and is gated by q_i (it cannot invent far-field energy); the cooling term keeps the correct sign. The dissipation Eq. (29) remains fully structure-preserving. The gains carry a $\Delta t/(\rho c_p)$ -type scale, initialized so $\text{softplus}(g_q) \approx 0.025$ (a per-step deposit of $q \Delta t$ yields the right $\sim 10^2$ K peak heating) and $\text{softplus}(g_c) \approx 10^{-3}$.

This enriched-source enthalpy GENERIC model has 1 375 238 parameters (+ 83 461 over the unconstrained backbone: the conductance MLP, the two source MLPs, and the global scales).

9 Training

9.1 Loss, optimizer, and training noise

The base objective is the mean squared error on the normalized increment $\Delta T = T^{t+1} - T^t$. We optimize with Adam (learning rate 10^{-3} , weight decay 10^{-4} , gradient-norm clipping at 1.0) and a

Algorithm 1 Push-forward training step (window of K step-aligned batches)

```
1: input: batches  $[B_0, \dots, B_{K-1}]$ , normalizer  $\mathcal{N}$ , model  $f_\theta$ 
2:  $\hat{T} \leftarrow \text{DYNAMICNOISE}(B_0.T, B_0.T_\infty)$  ▷ seed from window start
3:  $\mathcal{L} \leftarrow 0$ 
4: for  $k = 0$  to  $K - 1$  do
5:    $\mathbf{x} \leftarrow B_k.\mathbf{x}$  with temperature column  $\leftarrow \hat{T}$ 
6:    $\Delta T \leftarrow \mathcal{N}_y^{-1}(f_\theta(\text{standardize}(\mathbf{x}), B_k.\text{edges}))$ 
7:    $T_{\text{pred}} \leftarrow \hat{T} + \Delta T$ 
8:    $T_{\text{true}} \leftarrow B_k.T + B_k.y$ 
9:    $\mathcal{L} \leftarrow \mathcal{L} + \|(T_{\text{pred}} - T_{\text{true}})/\sigma_T\|^2/N$ 
10:   $\hat{T} \leftarrow T_{\text{pred}}$  ▷ feed forward; no detach  $\Rightarrow$  BPTT
11: end for
12:  $\mathcal{L} \leftarrow \mathcal{L}/K$ ;  backprop, clip, optimizer step
```

plateau learning-rate scheduler. To stabilize autoregressive rollouts we inject zero-mean Gaussian noise into the input temperature during training and correct the target accordingly (the standard MeshGraphNet noise of Pfaff et al. [1]):

$$\tilde{T}_i^t = T_i^t + \eta_i, \quad \eta_i \sim \mathcal{N}(0, \sigma^2); \quad \Delta T_i^{\text{target}} = T_i^{t+1} - \tilde{T}_i^t = y_i - \eta_i, \quad (33)$$

so the network learns to drive a perturbed state back onto the true trajectory. We use $\sigma = 5$ K. Noise is applied only to the training fold and only to the (raw) temperature column, before normalization.

9.2 Push-forward (multi-step) training

A recurring empirical lesson was that improving the single-step objective harder (lower training MSE, lower short-rollout validation RMSE) did *not* improve, and sometimes *worsened*, the long inference rollout: the single-step objective is simply not aligned with the 1526-step inference horizon. We therefore train with a *push-forward* objective: unroll the model K steps and backpropagate through the whole unroll (full backpropagation through time).

Concretely, the data loader yields windows of K consecutive same-simulation graphs $[g_t, \dots, g_{t+K-1}]$. We seed temperature from the (noised) true field at the window start and roll forward, feeding the model’s own prediction back as the next input’s temperature column, accumulating a scaled-temperature MSE over the window (Algorithm 1). $K = 1$ recovers the classic single-step objective. We use $K = 2$, which makes the operator contractive over two-step composition—enough, combined with the thermodynamic structure, to align with the long rollout while remaining cheap.

9.3 Validation metric and model selection

Validation uses the *full autoregressive rollout* RMSE on held-out simulations (not single-step error), averaged over the held-out fold; the best checkpoint is the one minimizing this metric. We additionally evaluate every checkpoint on the out-of-distribution spiral (below), because even the rollout validation does not perfectly track the very-long inference horizon.

9.4 Configuration and compute

The headline run uses $d = 128$, $L = 8$ message-passing steps, batch size 16, $K = 2$ push-forward, $\sigma = 5$ K noise, plateau scheduler, 80 epochs, simulation-level split (288 train / 72 val, validation

rollout capped at 20 simulations for cost). Training was performed on a single NVIDIA A100-80GB (~15 min/epoch, ~58 GB VRAM at this batch/ K); the processed-graph cache lives on a network volume so re-launches skip the build. Local evaluation/rollouts run on an RTX-class GPU.

10 Experiments and results

10.1 Out-of-distribution test: the Archimedean spiral

The held-out generalization probe is a constant-speed Archimedean-spiral weld $\gamma(\theta) = c + b\theta [\cos \theta, \sin \theta]$ on a square plate of side 0.2 m, 3 turns, travel speed 10 mm/s, net power ηP with $P = 2500$ W, $\eta = 0.8$. This trajectory is *never* seen in training (all training paths are straight/diagonal/sinusoid/arc), yet plate size and speed are interior to the training ranges (Section 3), so the spiral isolates trajectory generalization. The FEM ground truth has 4225 nodes and 1526 time steps (76.25 s)—an autoregressive horizon many times longer than any training window. The spiral additionally exercises genuine *re-heating* physics: successive spiral arms re-heat one another, a multi-pass effect absent from the single-pass training data.

10.2 Headline result

The enriched-source enthalpy GENERIC model trained with push-forward $K = 2$ attains:

- **Held-out single-pass validation rollout RMSE** that is *stable* across epochs and does not degrade (best 34.5 K at epoch 60), unlike the $K = 1$ variant which improved validation while drifting the spiral worse;
- **Spiral global RMSE** 33.2 K (best-validation checkpoint), about $2\times$ better than the best non-structure-preserving model obtained at any point in the study (65 K), and *stable* over all 1526 steps (bounded $\max |\text{err}|$, no divergence);
- **Maximum-principle respect**: minimum temperature stays within ~ 6 K of the physical floor (ambient/Robin bath), never cooling sub-ambient.

Figure 2 shows the five-probe thermal histories (FEM vs. surrogate) at radii spanning the melt pool to the cool corner. The shapes and arrival times of the passing peaks are captured; the one remaining flaw is a mild smoothing of the sharpest central peak (1713 K vs. FEM 1921 K), while the mid-field is essentially nailed (e.g. 908 K vs. 949 K).

10.3 Ablation: thermodynamic structure ON vs. OFF

To isolate the value of the structure, we train an otherwise identical *unconstrained* MeshGraphNet (GENERIC OFF, decoder active) with the same push-forward $K = 2$, the same split, budget, noise, and hyperparameters. The two models therefore differ *only* in the head, and their validation metrics are directly comparable. We then sweep every checkpoint (epochs 10–80) on the spiral and measure not just best-case RMSE but the *stability signals*: rollout-RMSE variance across checkpoints, the minimum temperature (maximum-principle violation), and whether the deployable best-validation checkpoint is also the best on the spiral.

Table 2 summarizes the ablation; Fig. 3 plots spiral-RMSE and minimum-temperature versus epoch for both models. The findings:

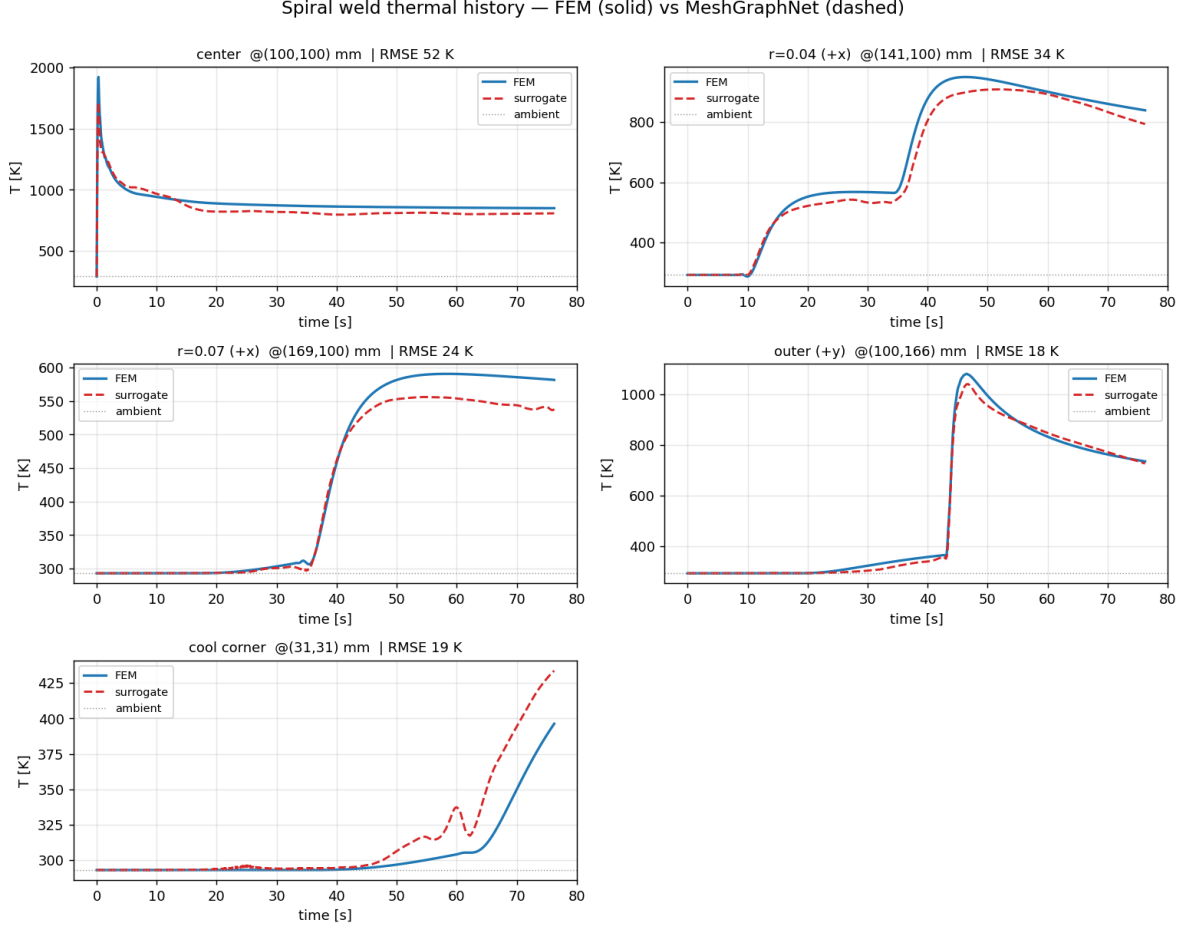


Figure 2: Spiral out-of-distribution test: FEM ground truth vs. the enriched-source enthalpy GENERIC surrogate (push-forward $K=2$, best-validation checkpoint), at five probe locations from the hot center to the cool corner. Global RMSE 33.2 K over 1526 autoregressive steps; stable, no divergence.

- **Maximum principle.** The structured model respects the physical floor ($\min T \approx 287$ K, ~ 6 K undershoot); the unconstrained model cools to 140 K to 190 K at *every* checkpoint—up to ~ 150 K of unphysical sub-ambient cooling.
- **Variance.** The structured model’s spiral RMSE sits in a tight band (33–61 K, $\sigma \approx 3.4$); the unconstrained model swings 38–111 K ($\sigma \approx 10.2$, $\sim 3\times$ larger).
- **Selectability.** For the structured model the best-validation checkpoint *is* the best spiral checkpoint (validation tracks the long rollout); for the unconstrained model the best-validation checkpoint gives a mediocre spiral (58.6 K) and the best spiral checkpoint cannot be selected by validation.
- **Held-out single-pass.** The structured model is also better on ordinary held-out rollouts (e.g. 8.1 K vs. 36.2 K on one validation simulation).

Table 2: Ablation: GENERIC enthalpy structure ON vs. OFF, both push-forward $K=2$, identical otherwise. The structured model respects the maximum principle, has $\sim 3\times$ lower spiral-RMSE variance, and a validation metric that tracks the spiral.

Configuration	Params	Val. RMSE (K)	Spiral RMSE @best-val (K)	Spiral range ep10–80 [σ] (K)	Min. T / undershoot (K)
ON (enthalpy GENERIC + enriched source)	1 375 238	34.5	33.2	33–61 [3.4]	287 (−6)
OFF (pure MeshGraphNet)	1 291 777	46.9	58.6	38–111 [10.2]	140 (−153)

Physical floor = 293 K (ambient = Robin bath = initial).

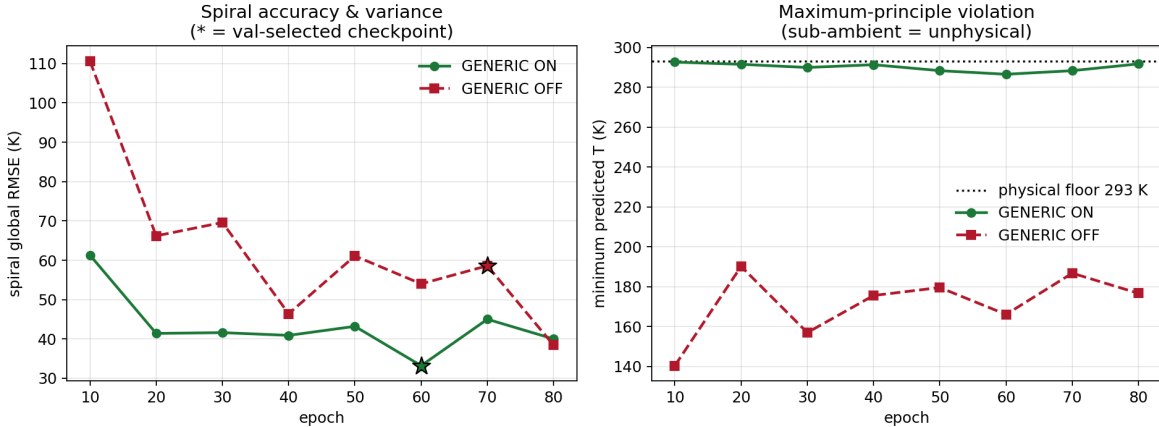


Figure 3: Ablation (both push-forward $K=2$). *Left*: spiral global RMSE vs. epoch; the structured model (ON) stays in a tight band and its best-validation checkpoint ($\star$) coincides with its best spiral, whereas the unconstrained model (OFF) swings $\sim 3\times$ more and its best-validation checkpoint is mediocre on the spiral. *Right*: minimum temperature vs. epoch; ON stays at the physical floor (293 K line), OFF cools sub-ambient at every checkpoint.

10.4 Held-out single-pass validation

Figures 5 and 6 show held-out single-pass rollouts (in-distribution trajectories, unseen geometry/process/boundary). The structured model is near-perfect in the far field (RMSE ~ 1 K, no spurious echo) and accurate at the peak; the residual error concentrates only in extreme dynamic-range outliers. These results—not the OOD spiral—constitute the core deployable claim: strong generalization across geometry, plate size, process parameters, and boundary conditions.

11 Discussion

What the structure buys. The honest framing of the ablation is not that the unconstrained network is catastrophic—at a lucky checkpoint it reaches a respectable number—but that it is *fragile*: you cannot select its good checkpoint by any deployable metric, it violates the maximum principle everywhere, and it varies $\sim 3\times$ more across training. The GENERIC structure instead provides *horizon-independent guarantees*: energy conservation and $dS/dt \geq 0$ hold at every single step regardless of rollout length, which is exactly what a 1526-step (or longer) deployment needs. This is a stronger and more defensible claim than a small RMSE improvement.

Spiral weld — FEM vs GENERIC ON (best ep60) vs GENERIC OFF (best ep70)

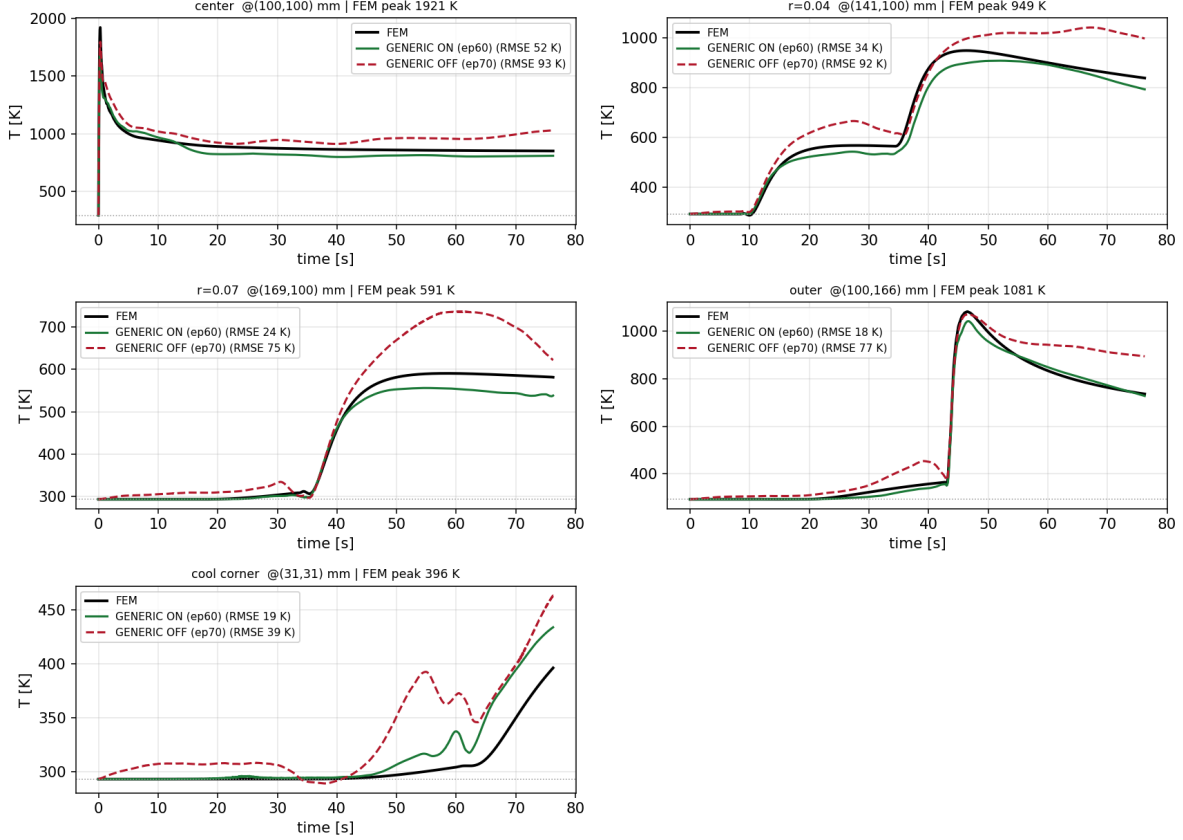


Figure 4: Three-way spiral thermal histories (five probes): FEM (black), GENERIC ON (green), pure MeshGraphNet OFF (red, dashed). The unconstrained model’s far-field and cool-corner behavior is visibly less faithful; per-curve RMSE in the legend.

Where the work is done. A caveat we make explicit: because the enriched source is a learned per-node MLP (gated by the Goldak field), one may ask whether the flexible source is “painting” the temperature rather than the structured dissipation doing the physics. The source is constrained (non-negative heating, gated by q , sign-correct cooling), but a principled next step is to replace the source MLP by a *named* temperature-dependent arc-efficiency $\eta(T)$ with few parameters: if a bounded physical mechanism still recovers the peak, the effect is real physics rather than black-box flexibility.

Limitations and future improvements. (i) **Exact energy conservation vs. the implemented uniform-volume reduction.** The first law in Proposition 1 holds on *any* mesh for the extensive-energy state $\mathcal{E}_i = h_i V_i$ Eq. (25). The deployed head uses the uniform-volume reduction $V_i \equiv V$, which is exact in the interior of the uniform tensor meshes used here and leaves only a small residual at boundary/corner nodes (tributary volumes $V/2$, $V/4$). On strongly non-uniform meshes—e.g. refined near the arc, as is common in production welding FEM—the unweighted conservation $\sum_i \Delta h_i = 0$ would no longer equal the physical $\sum_i V_i \Delta h_i = 0$. We emphasize that this does *not* affect the reported results: on the present uniform meshes the two formulations are numerically indistinguishable away from $\partial\Omega$, so the uniform-volume head is an excellent approximation,

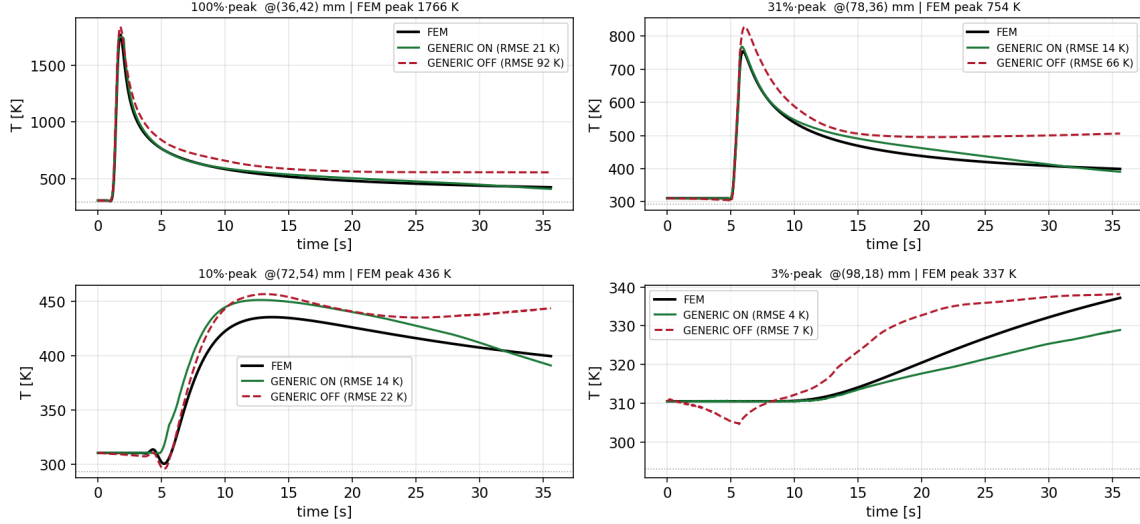


Figure 5: Held-out single-pass validation simulation (four peak-fraction probes): FEM (black), GENERIC ON (green), OFF (red, dashed). The structured model tracks all probes; global RMSE 8.1 K (ON) vs. 36.2 K (OFF).

and the maximum-principle/stability guarantees observed empirically are unaffected. Restoring *exact* conservation on arbitrary meshes requires only the cheap, learning-free volume weighting (the lumped FEM mass diagonal, already available); we leave the V_i -weighted formulation and its retraining on refined meshes as a straightforward future improvement. (ii) The model is 2D (thickness-integrated); a 3D extension follows the same graph machinery. (iii) The constant- ρc_p rescaling in Section 7.3 is exact only up to the curve $H(T)$ used; the curve is the FEM’s, so the surrogate inherits the FEM’s material model. (iv) The training corpus has a small soft-cap tail of very hot (>4000 K) simulations; tightening the vaporization cap and regenerating would clean it (the enthalpy head’s $H(T)$ table auto-syncs). (v) The OOD spiral’s central peak is still mildly smoothed; sharper-peak modeling is future work.

12 Related work

Learned mesh simulators. MeshGraphNets [1] and graph network simulators [2] established message passing on simulation meshes; our backbone is a configurable MeshGraphNet with the standard training-noise trick for rollout stability. *Structure-preserving learning.* The GENERIC framework [3, 4] has been used to learn thermodynamically consistent dynamics (e.g. GENERIC neural networks and structure-preserving operators) [8]; our contribution is to identify that imposing it on *temperature* is inconsistent under latent heat and to fix it with the classical *enthalpy method* [7], realized as a learned SPSP graph-Laplacian on a graph-network surrogate. *Welding/AM thermal surrogates* have used POD/ROM and CNN/GNN models; the novelty here is the combination of thermodynamic structure, phase-change-consistent energy bookkeeping, and push-forward training for long-horizon stability.

Held-out val sim_006 — FEM vs GENERIC ON vs OFF

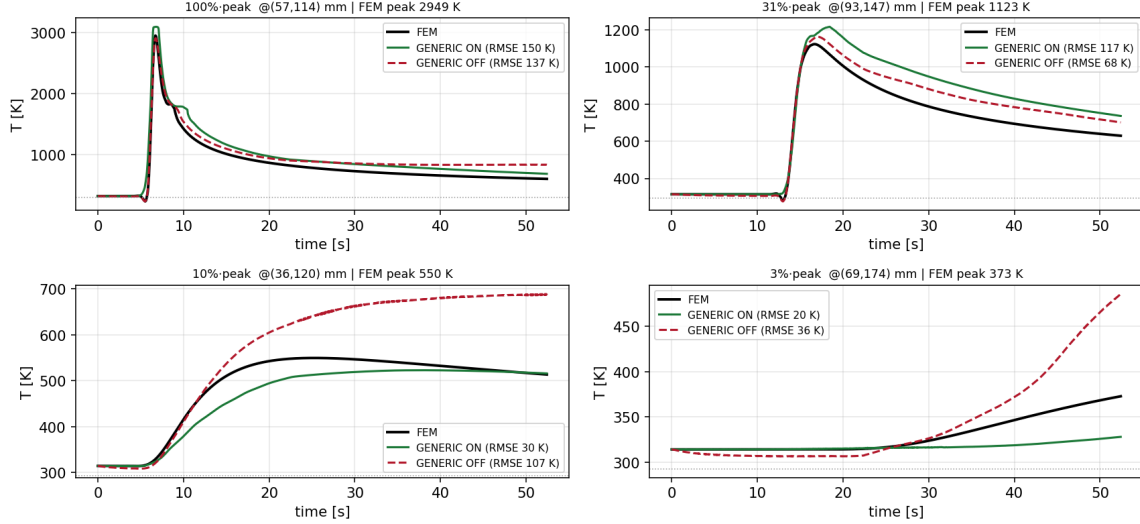


Figure 6: A second held-out validation simulation. Global RMSE 36.0 K (ON) vs. 63.4 K (OFF); the ON residual is concentrated at the sharp peak, the far field is near-exact.

13 Conclusion

We presented a MeshGraphNet surrogate for transient welding heat transfer that is *thermodynamically consistent by construction*. The key technical insight is that GENERIC structure must be imposed on *energy*, not temperature: with latent heat, conserving the sum of nodal temperatures is not conserving energy, and the mismatch is largest exactly at the melt pool. Carrying the volumetric enthalpy as the conserved state (the enthalpy method), evolving it with a learned SPSP graph-Laplacian, and recovering temperature through the latent-heat-aware curve restores genuine energy conservation and the second law through phase change. An enriched, non-conservative source recovers melt-pool peak heating without breaking the guarantees, and push-forward $K=2$ training aligns the objective with long autoregressive rollouts. On a held-out out-of-distribution spiral weld of 1526 steps the model reaches 33.2K global RMSE while respecting the maximum principle and remaining stable, and a matched ablation shows the structure delivers low-variance, selectable, physically admissible rollouts where the unconstrained network is accurate-but-fragile.

Reproducibility

The pipeline comprises: the FEM solver (`thermal_solver.py`: Goldak source, apparent heat capacity, θ -method + Picard), dataset generation (`generate_dataset.py`: stratified sampling and adaptive cooling), the graph builder (`graph_builder.py`: 12-d relative node features, 3-d edge features), the model (`meshgraphnet.py`: Encoder–Processor–Decoder plus the `EnthalpyGenericThermalHead`), and training (`train.py`: push-forward unroll, training noise, simulation-level split). The structure-preserving guarantees ($\sum_i \Delta H_i^{\text{diss}} = 0$, $dS/dt \geq 0$, latent-heat damping) are unit-tested. Normalization constants and the enthalpy table ride in the model `state_dict`, so checkpoints reproduce the exact physics.

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